

Dr Thomas Frost

📍 London, UK | ✉ thomas.frost@doctors.org.uk | 🌐 [tdgfrost](https://tdgfrost.com) | 📄 [drthomasfrost](https://drthomasfrost.com) | 👤 thomasfrost.io | 📖 [google scholar](https://scholar.google.com/citations?user=...)

PROFILE

Oxford-trained clinician and machine learning researcher, concluding a PhD in AI-enabled healthcare at UCL. I specialise in reinforcement learning for high-stakes clinical systems, with my research spotlighted on the [TalkRL podcast](#) alongside guests from DeepMind, OpenAI, and Meta. First-author publications on RL methods applied to healthcare, an open-source clinical RL dataset, and eight years practising medicine in the NHS.

EXPERIENCE

Doctoral Researcher, Machine Learning London, UK
Institute of Health Informatics, UCL *Sept 2022 — Present*

- Built an offline RL recommender system for insulin dose optimisation, predicted to reduce ICU mortality by 0.5–0.7%.
- Created and published [Insulin4RL](#), the first freely available dataset for RL using naturally-timed health data, containing 375,000 clinical decisions for 12,000+ patients.
- Developed overfitting-resistant mortality prediction models using temporal-difference learning, improving externally validated AUROC by ~14% over conventional supervised learning methods.
- Partnered with Microsoft and UCLH to help refine [FlowEHR](#), an MLOps project for deployment of clinical ML models.

Emergency Medicine Clinician London, UK
Homerton University Hospital *Sept 2021 — Present*

- Delivering frontline care in high-pressure environments, routinely making safety-critical decisions with incomplete data.
- Coordination of multidisciplinary teams for management of trauma and resuscitation for unwell patients.

Foundation Doctor; Junior Clinical Fellow in Emergency Medicine Edinburgh, UK
NHS Lothian *Aug 2018 — Jul 2021*

EDUCATION

University College London (UCL) London, UK
PhD in AI-enabled Healthcare *Sept 2022 — Nov 2026*

- **Thesis (submitted):** “Offline reinforcement learning for clinical data with natural timings”

University College London (UCL) London, UK
MRes in AI-enabled Healthcare *Sept 2021 — Sept 2022*

University of Oxford Oxford, UK
BM BCh Medicine; MA Medical Sciences (First Class) *Oct 2012 — Jul 2018*

TECHNICAL SKILLS

- **Languages & Core:** Expertise in Python, PyTorch, Git, Pandas/Polars/NumPy/SciPy.
- **Machine Learning:** Reinforcement learning, Causal inference, Time-series forecasting, Off-policy evaluation.
- **Infrastructure & Data:** AWS (EC2/S3), FlowEHR (TRE User/Validator), EHR time-series data cleaning.
- **Clinical AI:** Human-in-the-loop and LLM evaluation, Clinical safety metrics, Model interpretability.

SELECTED PUBLICATIONS

- Frost, T.** & Harris, S. (2026). Insulin4RL: Real-time insulin infusions for offline reinforcement learning. PhysioNet.
- Frost, T.**, Vaidya, H. & Harris, S. (2026). The hidden risks of temporal resampling in clinical reinforcement learning. Preprint arXiv:2602.06603.
- Frost, T.**, Li, K. & Harris, S. (2025). Robust real-time mortality prediction in the intensive care unit using temporal difference learning. In: Proceedings of the 4th Machine Learning for Health Symposium, PMLR, pp. 350–363.
- Ellershaw, S., Tomlinson, C., Burton, O.E., et al. (2024). Automated generation of hospital discharge summaries using clinical guidelines and large language models. In: AAAI 2024 Spring Symposium on Clinical Foundation Models.

AWARDS AND RECOGNITION

- **UKRI Studentship:** Full PhD stipend + research budget, for outstanding research potential.
- **UCL Centre of Digital Innovation Scholar:** £5,000 AWS research credits + cloud mentorship.
- **UCL Three-Minute Thesis:** University finalist, for communicating AI to non-technical audiences.
- **Lord Crewe Scholar (Oxford):** Awarded for academic distinction.